Project ID: R26-DS-11

Project Title: AI BASED EMERGENCY CALL AND SMS TRIAGE SYSTEM FOR DISASTER RESPONSE IN SRI LANKAN LOW RESOURCE LOCAL LANGUAGES

1. Introduction

1.1 Background

Sri Lanka is among the most disaster-prone countries in South Asia, receiving floods, landslides, cyclones, and droughts that cause harm to a significant number of people. In disaster scenarios, emergency call centers, run by organizations like the Disaster Management Center (DMC), face a challenge in handling many rescue calls at the same time made by distressed citizens in Sinhala, Tamil, English, or a combination of all three languages.

The current method of emergency communication is characterized by a first come first served system where human operators who have received training attend to every call, assess their urgency, gather information about the event and arrange the resources for dealing with the situation. This system works well during ordinary times, but it becomes a challenge when the number of calls increases.

The fact that there is multilingualism and a lack of resources in the communication setting of Sri Lanka presents another problem. Sinhala and Tamil have far fewer speech and natural language processing resources than English. The callers could change language mid-conversation, use local slang, be in a distressed or noisy situation, or text using Romanized and shortened SMS messages because of poor connectivity.

Current developments in the fields of speech processing, natural language processing, machine learning, and real-time communications technology make it feasible to develop an automated triage system. This system will help with urgency assessment, information extraction, adaptation, deduplication of incidents, and situational awareness for emergency call handlers.

1.2 Research Problem

The Sri Lankan emergency response network does not have an intelligent system that is capable of analyzing multilingual emergency calls and texts, determining the urgency of such emergencies, gathering any missing data, merging duplicates, and offering prioritized emergency information to disaster managers. The current system analyzes incoming data on a serial basis, provides partial capability of handling the Sinhala and Tamil languages, and is poorly equipped to handle low-bandwidth SMS.

The research problem is therefore formulated as follows:

> *"How can an AI-powered emergency communication triage system be developed to automatically analyze multilingual emergency calls and SMS messages in Sri Lankan low-resource languages, assess incident urgency using prosodic and conversational cues, extract structured incident information, and present fused situational intelligence to disaster management operators in real time?"*

1.3 Objectives

IT Number	Objective	Objective No.
IT22644480	Multimodal Urgency Detection in Emergency Calls Using Prosodic and Conversational Cues	1
IT22129758	Multilingual Emergency Conversational AI System for Rapid Clarification	2
IT22263476	Low-Bandwidth SMS Emergency Intelligence for Disaster Response	3
IT22504418	Multi-Source Urgency-Based Incident Fusion, Unified Incident Records and Real-Time Operator Dashboard	4

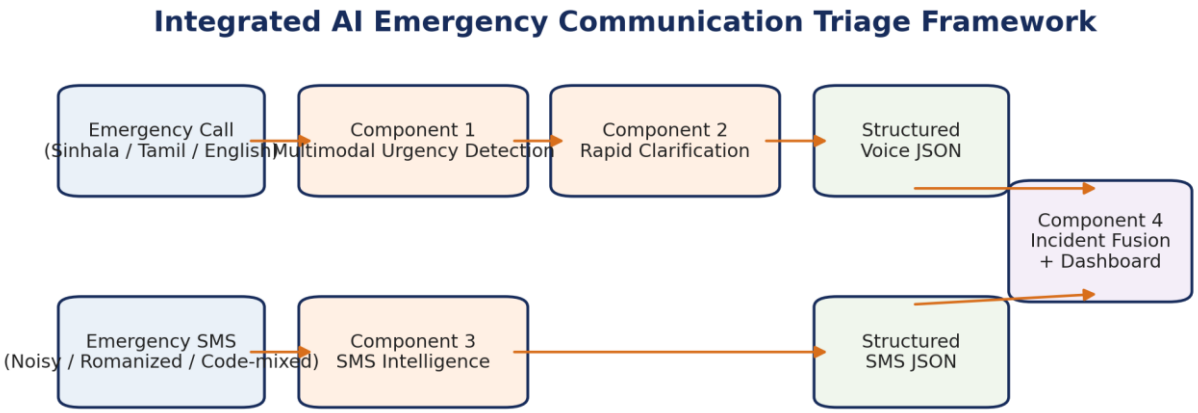


Figure 1. Integrated relationship between the four research components.

2. Data Exploration

2.1 Data Collection

2.1.1 Component 01: Multimodal Urgency Detection Using Prosodic and Conversational Cues

The data collection for Component 01 is done from two main sources. The first source includes anonymous disaster-related emergency calls that have been recorded using the help of Disaster Management Centre of Sri Lanka. These emergency calls are multilingual calls made in Sinhala and Tamil language by the callers.

In order to overcome the problem of lack of real world emergency speech data, a large dataset of simulated calls is also created. Multilingual emergencies are scripted at High, Medium, and Low urgency and performed by volunteer callers in Sinhala, Tamil, and English languages.

openSMILE, which utilizes GeMAPS v2, is applied for extracting prosodic and paralinguistic features. Multilingual transcription is provided by Whisper. Audio-based features are the pitch range and mean, loudness, speaking rate, voice quality, and pause patterns; transcription-based features include the presence of an emergency keyword, repetition, lack of complete sentences, and emotional elements.

2.1.2 Component 02: Multilingual Emergency Conversational AI System for Rapid Clarification

Component 02 consists of structured multilingual emergency dialogues using realistic floods, landslides, fires, medical emergencies, and structural collapse situations. These dialogues deliberately leave out information like exact location, number of people affected, injured status, and seriousness of the situation so that the system learns where it needs to ask for clarifications.

There are simulated caller-agent dialogues in Sinhala, Tamil, English, and code-switched dialogues in the dataset. Intent patterns and entity tagging in emergency dialogues and South Asian conversational corpora are referred to as existing datasets. Twilio takes the live calls and routes it into the system, while Whisper generates the transcription needed by the downstream dialogue pipeline.

The clarification phase is structured in a modular process of language and code-mix detection, entity recognition, dialogue act and relevance recognition, information gap detection, clarification strategy selection, question generation, confidence assessment, and JSON structuring.

2.1.3 Component 03: Low-Bandwidth SMS Emergency Intelligence for Disaster Response

Component 03 draws upon open-source disaster messaging corpora like CrisisNLP and HumAID, while the main SMS corpus is created artificially to represent emergencies in Sinhala, Tamil, English, and Roman script code-mixing. These messages contain spelling mistakes, shortened words, incomplete fields, inconsistencies in Roman script usage, etc.

Twilio SMS is used for message processing. Language/script identification is done using langdetect and fastText. Text Normalization uses rule-based Normalization Table, spaCy, NLTK, and sequence-based approaches. Named Entity Recognition recognizes locations, number of persons involved, types of incidents, and resource requirements. Urgency is predicted by scikit-learn classifiers.

2.1.4 Component 04: Multi-Source Urgency-Based Incident Fusion and Real-Time Operator Dashboard

In Component 04, structured JSON files are received from the voice and SMS pipelines as opposed to harvesting raw field data. In order to test and validate the fusion algorithm, 20 multilingual disaster scenarios will be generated, creating roughly 300 reports with Unified Incident Record ground truth labels.

Multilingual gazetteer for Sri Lanka containing over 5,000 variants of location names in Sinhala Unicode, Tamil Unicode, romanized Sinhala, romanized Tamil, English, and landmarked forms is created. Gazetteer variants are geocoded using OpenStreetMap and government boundaries data.

LaBSE gives us 768-dimensional cross-lingual semantic embeddings. PostgreSQL with PostGIS holds the Unified Incident Records and allows us to index the data spatially, and Redis cache embedding centroids to reduce the clustering latency when receiving a large amount of reports.

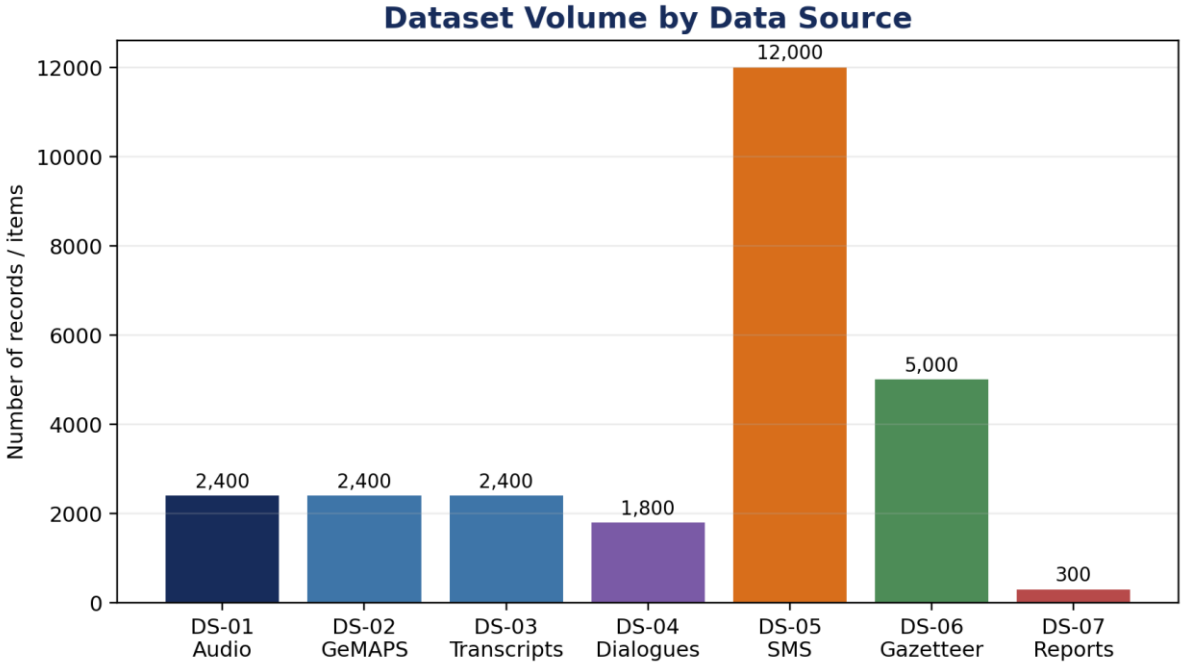


Figure 2. Relative volume of the seven main datasets used in the project.

2.2 Dataset Description

Data Source	Description	Resource	Size	Key Attributes
DS-01 (Prosodic Audio Dataset)	Multilingual emergency call recordings, real and simulated, used for acoustic feature extraction and urgency labelling. Covers Sinhala, Tamil, and English.	DMC Sri Lanka + Simulated	~2,400 audio segments; ~18 hours; ~6.8 GB	Audio clip; language; urgency level; duration; speaker ID
DS-02 (GeMAPS Feature Dataset)	Eighty-eight-dimensional GeMAPS prosodic feature vectors extracted from each audio segment using openSMILE.	openSMILE / GeMAPS v2	2,400 feature vectors; ~4.1 MB	Pitch mean/range; speech rate; HNR; loudness; jitter; shimmer; pause ratio
DS-03 (Whisper Transcription Dataset)	Speech-to-text transcriptions of the emergency audio in Sinhala, Tamil, and English.	OpenAI Whisper	2,400 transcripts; ~3.2 MB	Transcript text; detected language; confidence score; word timestamps

These first three datasets feed into the voice processing path. DS-01 retains the original sound wave, DS-02 has pre-defined length prosody features to model urgency, and DS-03 offers multi-lingual text for conversational cueing.

Data Source	Description	Resource	Size	Key Attributes
DS-04 (Conversational Dialogue Dataset)	Scripted multilingual emergency dialogues between callers and the clarification agent, annotated for missing fields and final structured records.	Simulated / SLIIT Research	1,800 dialogue sessions; ~2.8 MB	Caller language; missing fields; extracted entities; clarification turns; final record
DS-05 (Emergency SMS Dataset)	Synthetic emergency SMS messages in Sinhala, Tamil, English, and Romanized code-mixed forms under low-bandwidth conditions.	Simulated + CrisisNLP	12,000 SMS records; ~1.9 MB	Raw text; language; noise level; urgency class; entities present
DS-06 (Sri Lanka Location Gazetteer)	Multilingual place-name database mapping script and landmark variants to geographic coordinates.	OSM + Government Data	5,000+ location variants; ~8.4 MB	Place name; script type; latitude; longitude; district; GN division

DS-04 supports adaptive clarification research, DS-05 provides the core low-bandwidth SMS corpus, and DS-06 acts as a shared geographic resource for SMS interpretation and cross-source incident fusion.

Data Source	Description	Resource	Size	Key Attributes
DS-07 (Fusion Scenario Dataset)	Scripted multi-source disaster scenarios with ground-truth Unified Incident Record labels for evaluation of fusion, conflict detection, and deduplication.	SLIIT Research / DMC	20 scenarios; ~300 reports; ~5.2 MB	Report ID; source; urgency; raw location; timestamp; ground-truth UIR ID

The fusion scenario dataset is smaller than the component-level training datasets because each record is a multi-source validation report linked to a known incident cluster. Its purpose is to measure whether the system merges true duplicates while keeping separate incidents distinct.

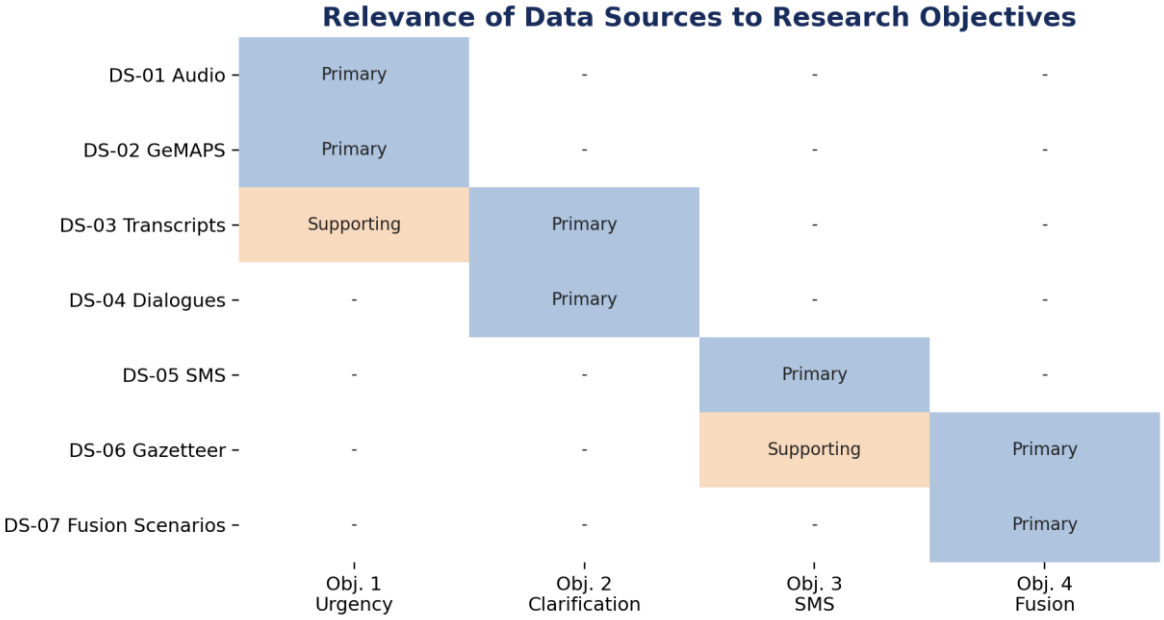


Figure 3. Primary and supporting roles of the project datasets.

2.3 Suitability Analysis

2.3.1 Relevance to Individual Research Objectives

Data Source	Objective 1: Urgency	Objective 2: Clarification	Objective 3: SMS	Objective 4: Fusion
DS-01 Prosodic Audio	Primary - raw audio input for acoustic feature extraction and urgency-model training.			
DS-02 GeMAPS Features	Primary - fixed 88-dimensional vectors supplied to Random Forest, SVM, and Neural Network classifiers.			
DS-03 Whisper Transcripts	Supporting - conversational cues from transcribed speech.	Primary - caller utterance text for language, relevance, dialogue-act, and entity extraction.		
DS-04 Dialogue Dataset		Primary - trains and evaluates gap detection, question decision, multilingual clarification, and final-record completion.		

The voice branch depends on complementary acoustic and textual information. The prosodic dataset is suitable for urgency estimation, while the transcript and dialogue datasets are necessary for understanding incident content and deciding how the system should respond to each caller turn.

Data Source	Objective 1: Urgency	Objective 2: Clarification	Objective 3: SMS	Objective 4: Fusion
DS-05 SMS Dataset			Primary - trains language detection, normalization, entity extraction, completeness analysis, and urgency classification.	
DS-06 Gazetteer			Supporting - resolves locations extracted from SMS messages.	Primary - normalizes cross-lingual and informal locations before clustering.
DS-07 Fusion Scenarios				Primary - evaluates deduplication precision, recall, conflict handling, Unified Incident Record quality, and dashboard latency.

The SMS and fusion datasets are also strongly aligned with their objectives. The gazetteer is especially important because it creates a shared geographic representation across Sinhala, Tamil, English, Romanized, and landmark descriptions, enabling both SMS interpretation and incident fusion.

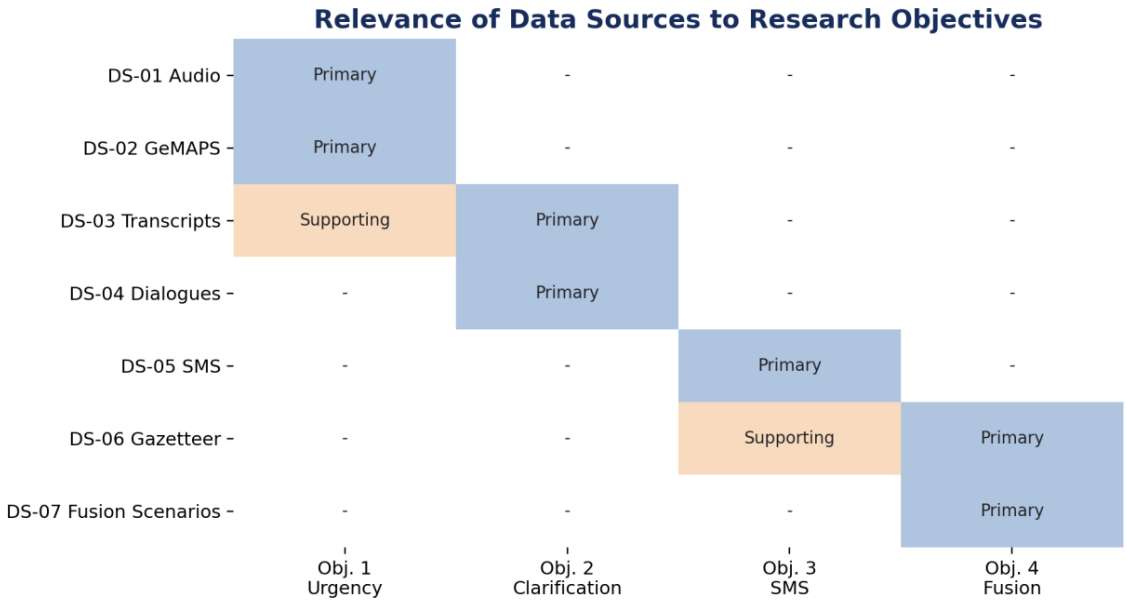


Figure 4. Visual summary of data suitability across the four objectives.

3. Methodology

3.1 Data Preprocessing

Objective 1 - Multimodal Urgency Detection

Data Source	Data Cleaning	Noise Reduction	Feature Extraction	Normalization	Encoding	Augmentation	Scaling
DS-01 Audio	Yes	Yes	Yes	Yes	Yes	Yes	No
DS-02 GeMAPS	Yes	No	No	Yes	No	No	Yes
DS-03 Transcripts	Yes	No	Yes	No	Yes	No	No

Audio preprocessing removes unusable segments, controls background noise, standardizes sampling properties, and supports augmentation of underrepresented urgency-language combinations. GeMAPS vectors are cleaned, normalized, and scaled before modelling. Transcripts are normalized for multilingual NLP while preserving emergency-critical wording.

Objective 2 - Multilingual Emergency Conversational AI

Data Source	Data Cleaning	Language Detection	Entity Extraction	Normalization	Encoding	Missing Data	Augmentation
DS-03 Transcripts	Yes	Yes	Yes	Yes	Yes	Yes	No
DS-04 Dialogues	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Dialogue preprocessing preserves turn order and caller language, aligns extracted entities with ground-truth fields, marks missing or untrusted information, and augments low-frequency scenarios. This supports a dialogue state representation rather than isolated one-turn question generation.

Objective 3 - Low-Bandwidth SMS Intelligence

Data Source	Data Cleaning	Text Normalization	Entity Extraction	Urgency Classification	Encoding	Missing Data	Augmentation
DS-05 SMS	Yes	Yes	Yes	Yes	Yes	Yes	Yes
DS-06 Gazetteer	Yes	No	Yes	No	Yes	No	No

SMS preprocessing corrects common abbreviations, phonetic spellings, character substitutions, Romanized forms, and truncation patterns. The process retains original text alongside normalized text so that operators can review both forms when confidence is low.

Objective 4 - Incident Fusion and Dashboard

Data Source	Normalization	Location Resolution	Embedding	Clustering	Conflict Detection	Deduplication	Scaling
DS-06 Gazetteer	Yes	Yes	No	No	No	No	No
DS-07 Fusion Scenarios	Yes	Yes	Yes	Yes	Yes	Yes	No

Fusion preprocessing standardizes incident type, urgency, source, time, and geographic fields before similarity calculation. Ground-truth cluster labels are retained only for evaluation and are not used as input features, preventing leakage into the clustering decision.

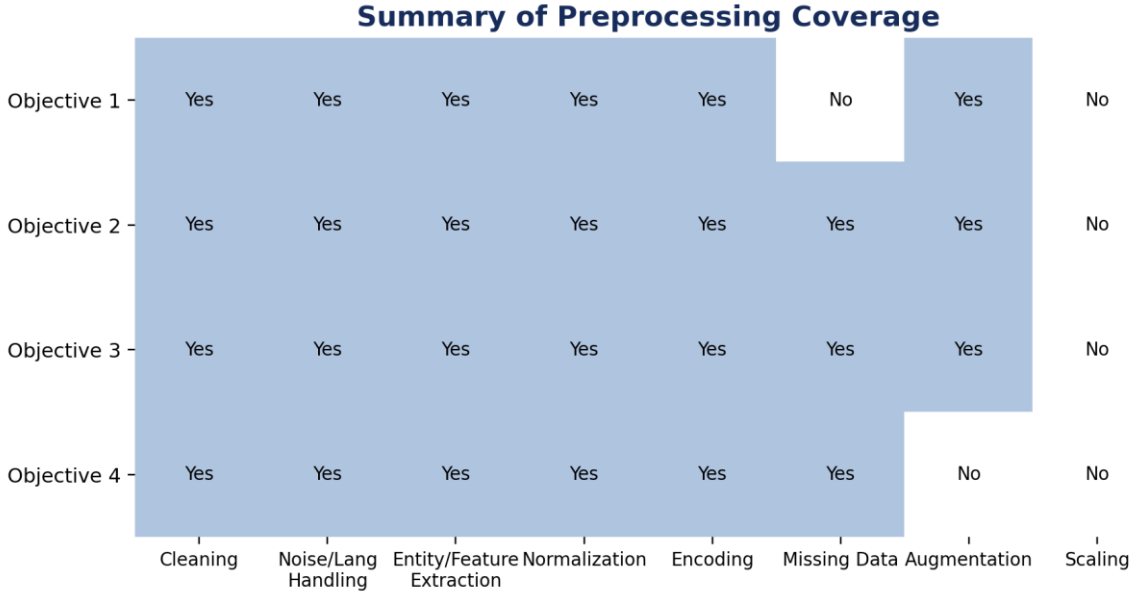


Figure 5. Coverage of the major preprocessing operations.

3.2 Scalability

Objective 1 - Multimodal Urgency Detection

The prosody database comprises roughly 2,400 audio clips. GeMAPS generates a fixed-length 88-dimensional feature space independent of audio length, which makes inference computation feasible. The use of a streaming framework allows processing via each call using Twilio, and extra DMC audio data may be added as needed to increase coverage for disasters, local accents, and different speakers.

Objective 2 - Multilingual Emergency Conversational AI

The dialogue corpus consists of 1,800 sessions from three languages and various information-gap settings. The clarification system architecture is designed as follows: language identification, relevance and dialogue act recognition, entity recognition, dialogue state, question policy, language generation, and safety checking can be individually scaled up or substituted. Parallel webhook workers allow for multiple simultaneous callers, whereas a max-turn policy prevents uncontrolled dialogue expansion.

Objective 3 - Low-Bandwidth SMS Intelligence

The SMS dataset containing 12,000 SMS messages is enough to build initial classification and extraction models. Since SMS messages are short, the pipeline remains relatively light. FastAPI can use several workers for handling spikes in messages, and the gazetteer may have more aliases added without retraining the urgency or extraction models.

Objective 4 - Incident Fusion and Dashboard

Fusion uses LaBSE embeddings, Redis-cached centroids, and PostGIS spatial indexing in order to handle many reports. The system decouples ingestion, processing, storage, and dashboard updates via queues, thus enabling independent scaling of each component. The design is built around eight seconds of end-to-end latency target for simultaneous ingestion.

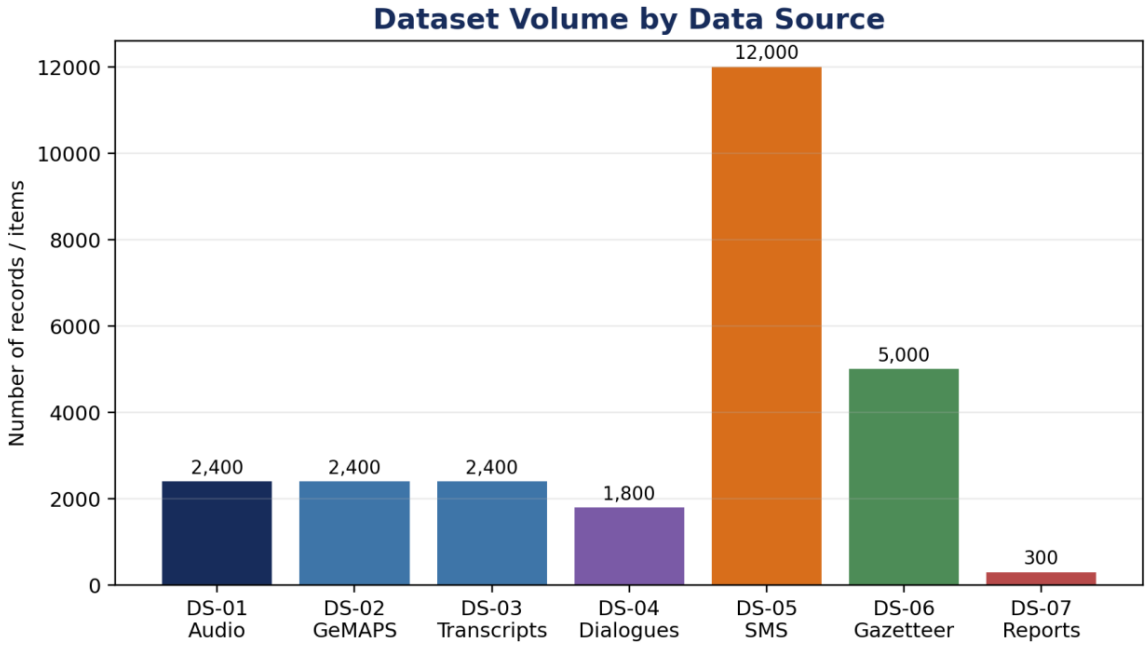


Figure 6. Dataset volumes relevant to scalability planning.

3.3 Feature Extraction

Objective 1 - Multimodal Urgency Detection

The feature extraction for Component 01 is done using both acoustic and linguistic data. openSMILE with GeMAPS v2 extracts 88 prosodic features including the statistics of pitch, loudness, speech rate, voice quality measures, proportion of pause, and spectral properties. The features are extracted using an analysis window of 25 ms and hop size of 10 ms.

In the Whisper transcript, emergency keyword density, repetition count, incomplete sentences, exclamations/ questions density, and semantic urgency score are extracted. The acoustic and transcript features are then normalized and combined into one vector so that critical event descriptions which are calm in nature are not evaluated just by tone of voice.

- Pitch and energy features: F0 mean, F0 range, loudness and loudness variation.
- Voice-quality features: HNR, jitter, shimmer and spectral descriptors.
- Temporal speech features: speaking rate, pauses and repeated phrases.
- Conversational features: emergency lexicon density, sentence completeness and semantic urgency.

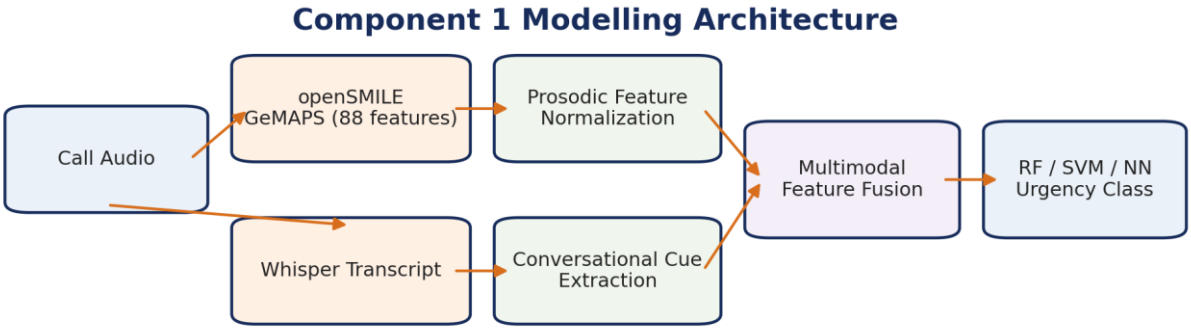


Figure 7. Multimodal feature extraction and urgency-classification architecture.

Objective 2 - Multilingual Emergency Conversational AI

Component 02 gathers structured incident information and dialogue control features for every caller turn. The mandatory information schema contains the following components: location, type of incident, number of people involved, injury status, entrapment status, seriousness parameters, and resources needed. Every extracted parameter is accompanied by the corresponding confidence and trust level because it may be required to confirm rather than accept the weak or conflicting information.

Dialogue state contains such pieces of information as relevance, dialogue act, questions asked before, failed attempts to clarify, caller's language, urgency context, and turn budget left. Such information enables clarification policy to make decisions about asking for information, confirmation, redirecting off-topic statements, softening the language for stressed caller, or stopping information gathering.

- Location entities: district, GN division, exact address, area and landmark descriptions.
- Incident entities: flood, landslide, medical emergency, fire and structural collapse.
- Human-impact entities: people count, injured, trapped, dead or missing.
- Dialogue-control features: relevance, dialogue act, confidence, missing-field priority and question history.

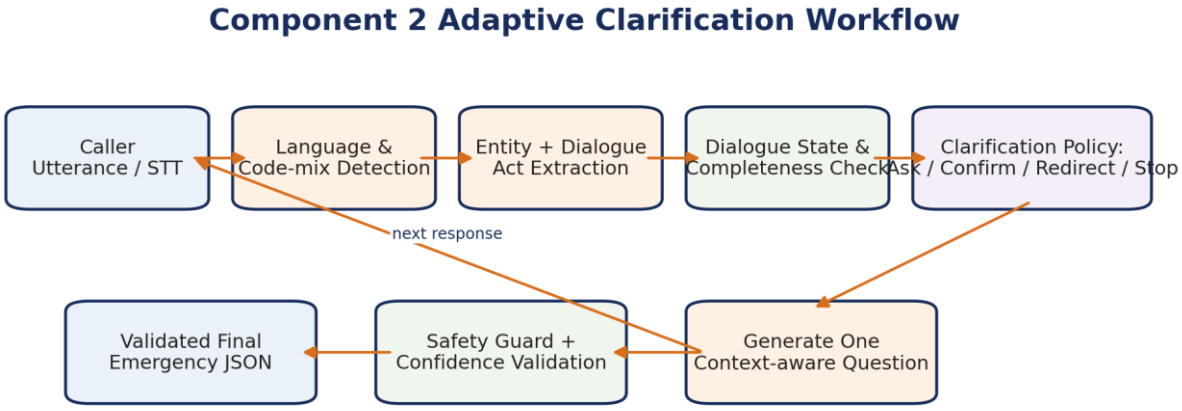


Figure 8. Adaptive clarification loop using dialogue state and a question-decision policy.

Objective 3 - Low-Bandwidth SMS Intelligence

SMS feature extraction begins with language and script detection. Normalization then corrects abbreviations, phonetic spellings, character substitutions, and Romanized Sinhala or Tamil forms. Named Entity Recognition extracts location, person count, disaster type, severity, and resource requirements from the normalized message.

Urgency features include emergency keyword score, message length, entity-completeness ratio, capitalization ratio, and punctuation density. These features are supplied to multi-class classifiers that produce Critical, High, Medium, or Low urgency labels. Incomplete records are flagged for a follow-up SMS or operator review.

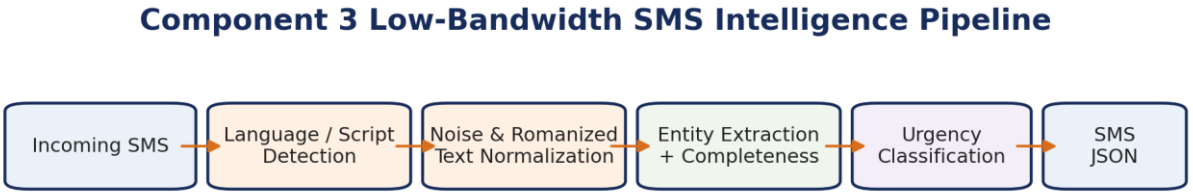


Figure 9. Sequential feature extraction for noisy multilingual SMS messages.

Objective 4 - Multi-Source Incident Fusion

The fusion module works on the basis of structured JSON records. Location is resolved on an exact match, fuzzy match based on RapidFuzz Levenshtein threshold of 85% and landmarks-based match. Resolved coordinates along with incident content and time help identify if two reports refer to the same event.

LaBSE embeds each incident content into 768 dimensional semantic vector. The distance measure includes semantic cosine distance with weight 0.5, geographical distance rescaled by 2 km threshold with weight 0.3, and temporal distance rescaled by 30 min threshold with weight 0.2.

DBSCAN eps parameter is chosen as 0.15 for critical incidents, 0.25 for high incidents, 0.3 for medium incidents, and 0.4 for low incidents.

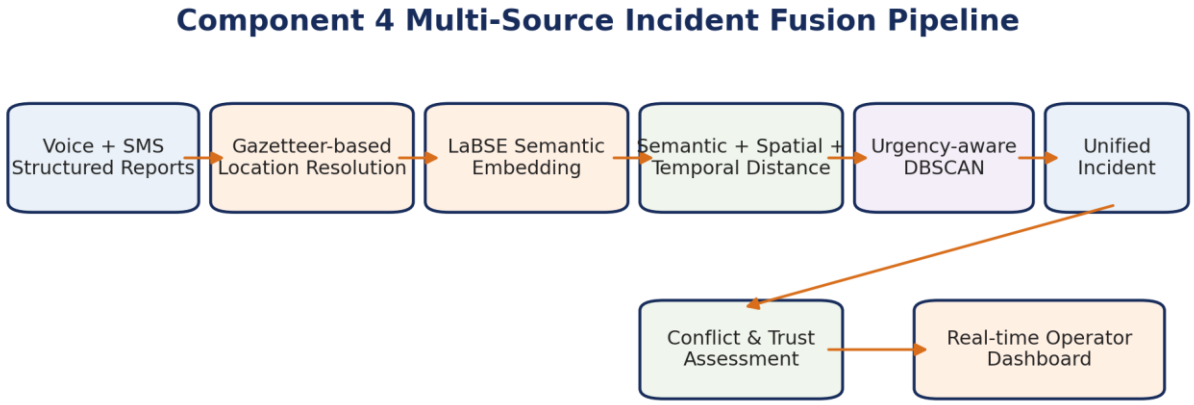


Figure 10. Cross-source location, semantic, temporal, and urgency-aware incident fusion.

4. Modelling and Results

Objective 1 - Multimodal Urgency Detection Using Prosodic and Conversational Cues

Three architectures are evaluated on the multimodal feature set. A Random Forest with 200 estimators serves as the primary interpretable model. An RBF-kernel Support Vector Machine provides a strong nonlinear comparison. A three-layer Neural Network with 256, 128, and 64 hidden neurons, ReLU activation, and dropout of 0.3 models interactions between acoustic and linguistic features.

The data are divided using a 70/15/15 train, validation, and test split stratified by urgency and language. Class weights address the lower frequency of high-urgency cases. Evaluation prioritizes recall for high-urgency calls because missing a critical incident is more harmful than producing an additional false alert.

Model	Role in Evaluation	Strength	Main Limitation
Random Forest (200 trees)	Primary urgency classifier and feature interpretation	Robust with moderate data; supports feature importance	May not fully model complex temporal interactions
SVM with RBF kernel	Nonlinear decision-boundary comparison	Strong performance with scaled fixed-length features	Less transparent and slower at large scale
Neural Network (256-128-64)	Learns nonlinear multimodal interactions	Flexible representation of combined prosodic and linguistic cues	Requires careful regularization and calibration

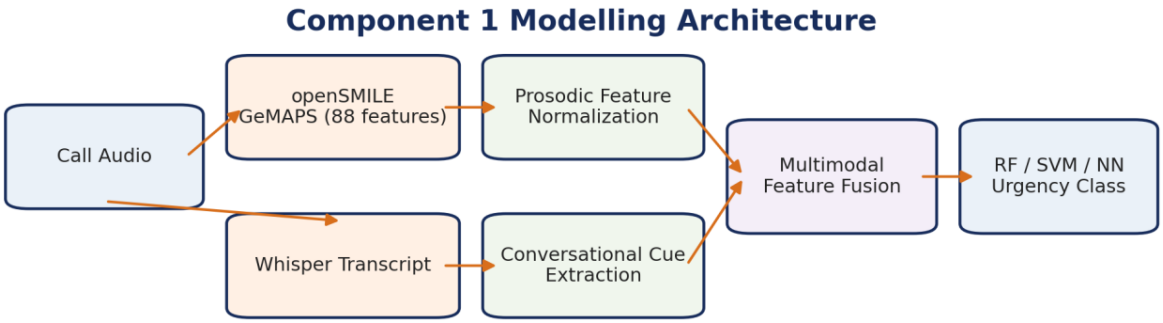


Figure 11. Component 1 training and inference flow.

Preliminary analysis indicates that pitch range and speech rate separate High and Low urgency effectively, while the Medium-to-High boundary needs linguistic context. Calm callers describing critical situations are a central safety challenge, which supports the use of multimodal fusion rather than prosodic information alone.

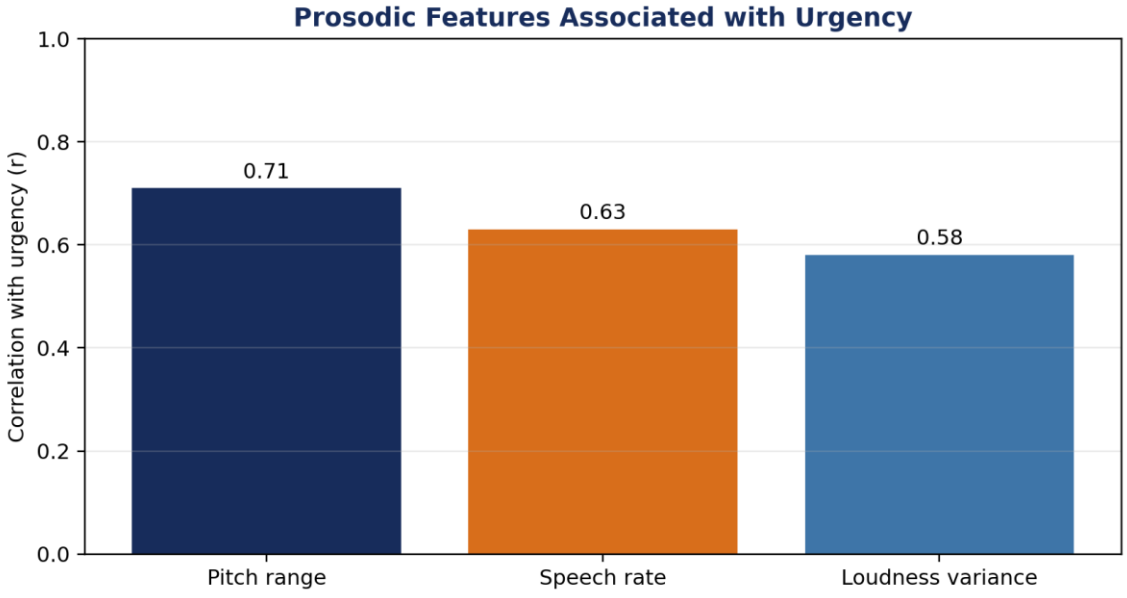


Figure 12. Reported correlations between urgency and key prosodic features.

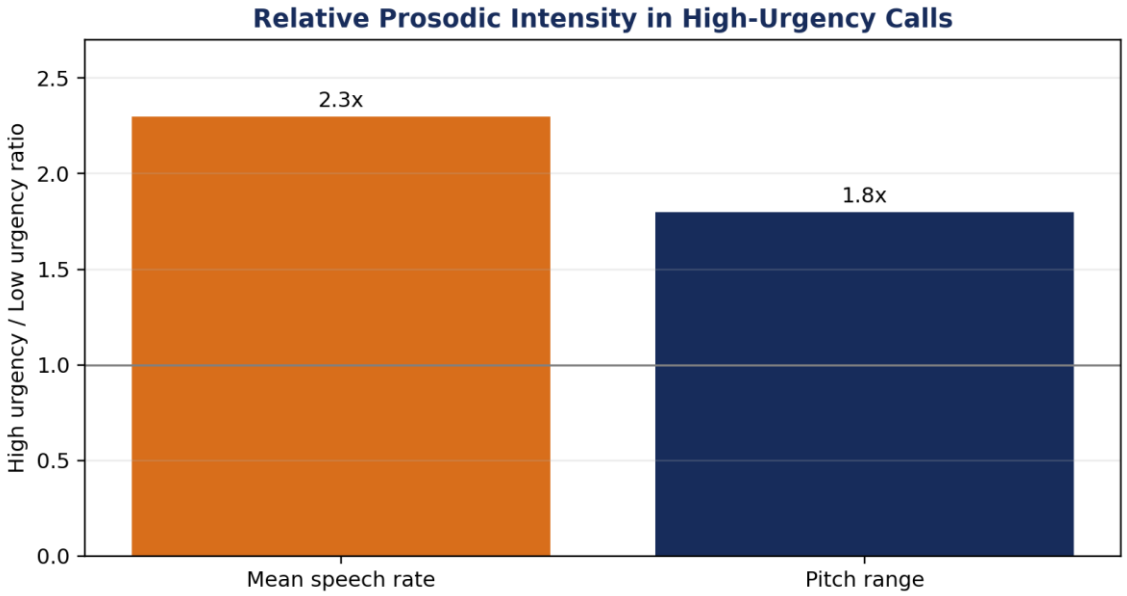


Figure 13. Relative speech-rate and pitch-range changes in high-urgency calls.

Objective 2 - Multilingual Emergency Conversational AI System

The clarification engine is designed to be an iterative dialogue system with structured output validation. Every transcript is analyzed for language, relevance, dialogue act, incident entities, confidence, and missing fields. Afterward, the policy makes its decision on what the next action should be, which is realized by the question generator in the caller's language.

This goes on until either all mandatory fields are filled out or until there are up to five turns of clarification. Anything with less than 0.7 confidence level is verified instead of just being accepted. Any off-topic or non-informative statement can be resolved using emergency domain relevance and dialogue-act determination, thus preventing the system from asking the same question over and over again no matter what the caller says.

Policy Action	When It Is Used	Expected System Behaviour
ASK	A high-priority required field is missing	Ask one concise, context-aware question in the caller language.
CONFIRM	An extracted field is weak, contradictory, or below the confidence threshold	Repeat the suspected value and request confirmation.
REDIRECT	The caller provides an off-topic or non-informative response	Acknowledge briefly and return to the unresolved emergency field.
SOFTEN	The caller shows emotional distress	Use empathetic wording and reduce question complexity.
STOP / FINALIZE	Required information is sufficient or the turn limit is reached	Produce the validated emergency JSON and flag remaining uncertainty.

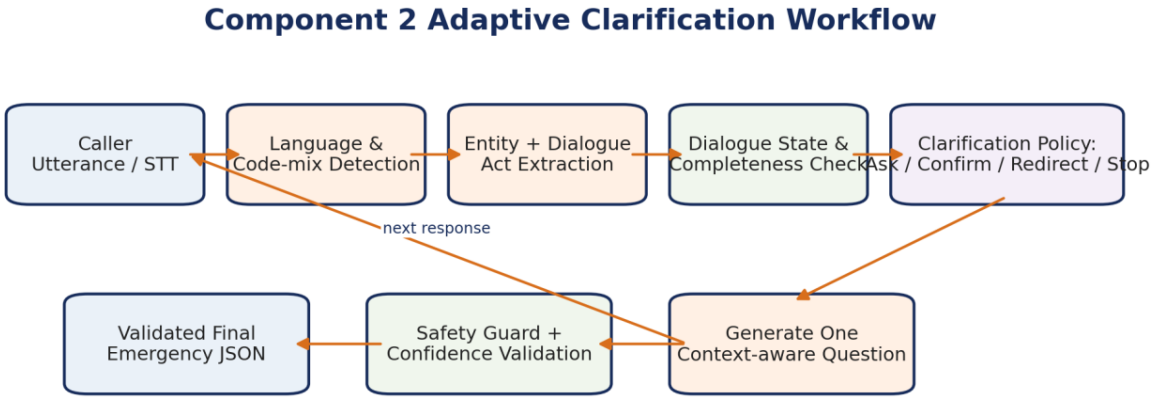


Figure 14. Clarification policy and validation cycle.

The final output is a validated structured emergency record. The record contains the caller language, incident type, location, human-impact indicators, urgency context, resource need, source confidence, unresolved fields, and conversation history required for audit and operator review.

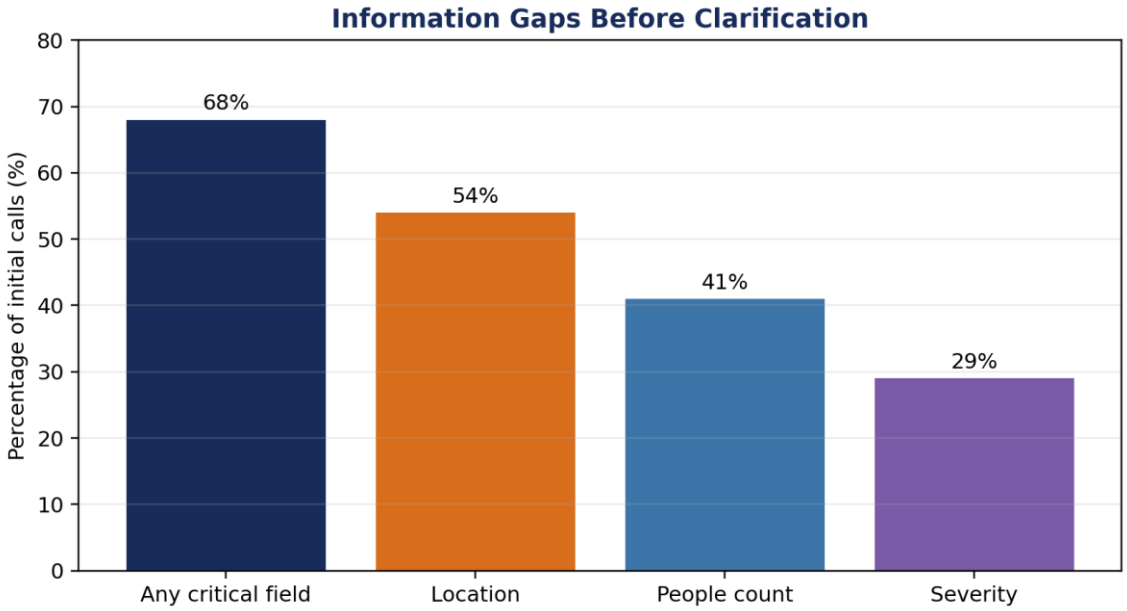


Figure 15. Frequency of critical information gaps in the dialogue dataset.

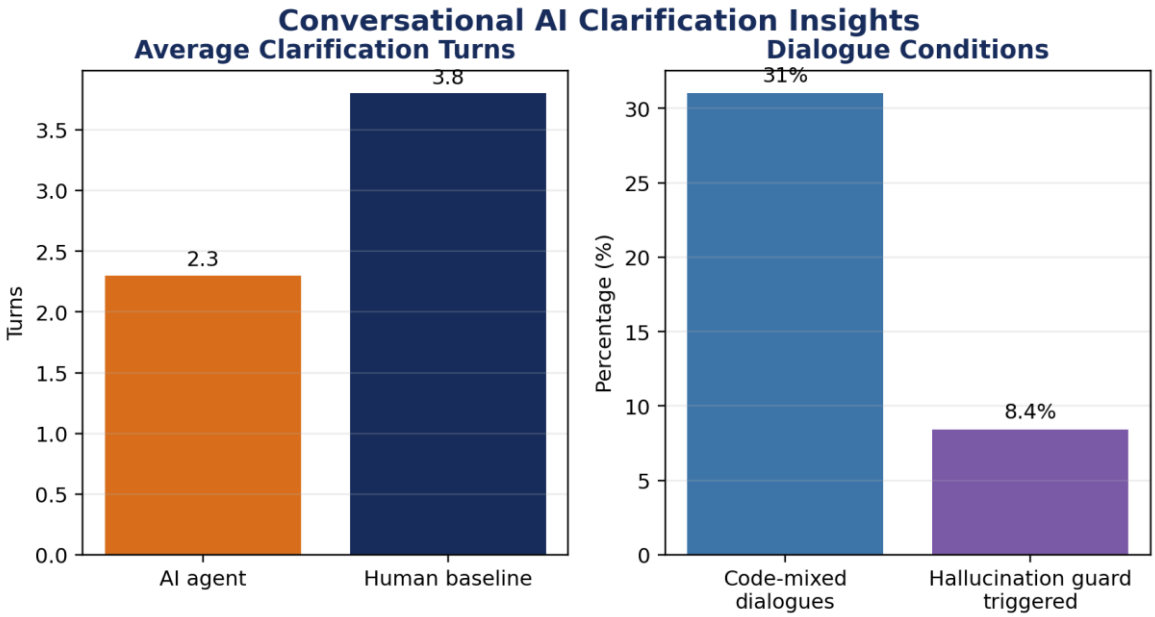


Figure 16. Clarification efficiency, code-mixing, and hallucination-guard activation.

Objective 3 - Low-Bandwidth SMS Intelligence

The SMS pipeline is evaluated on the synthetic multilingual test set. Language detection performs best on clean text and degrades on strongly Romanized code-mixed messages, requiring a character n-gram fallback. Normalization is applied before entity extraction to reduce spelling and script variation.

Random Forest is used as the baseline urgency classifier and SVM provides comparison. Evaluation is performed across complete messages, messages missing location, and messages with several missing fields. The completeness checker determines whether a follow-up message is required and produces a structured JSON record containing incident type, resolved location, people count, urgency level, and resource need.

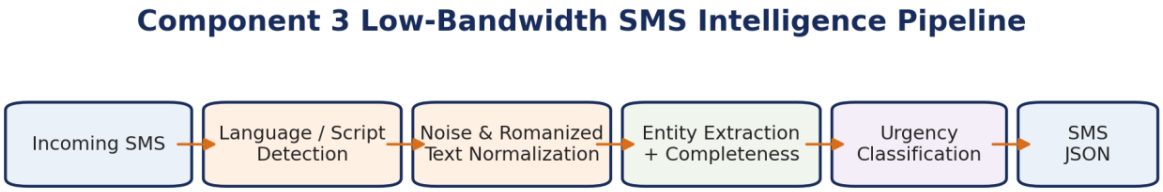


Figure 17. Component 3 modelling workflow.

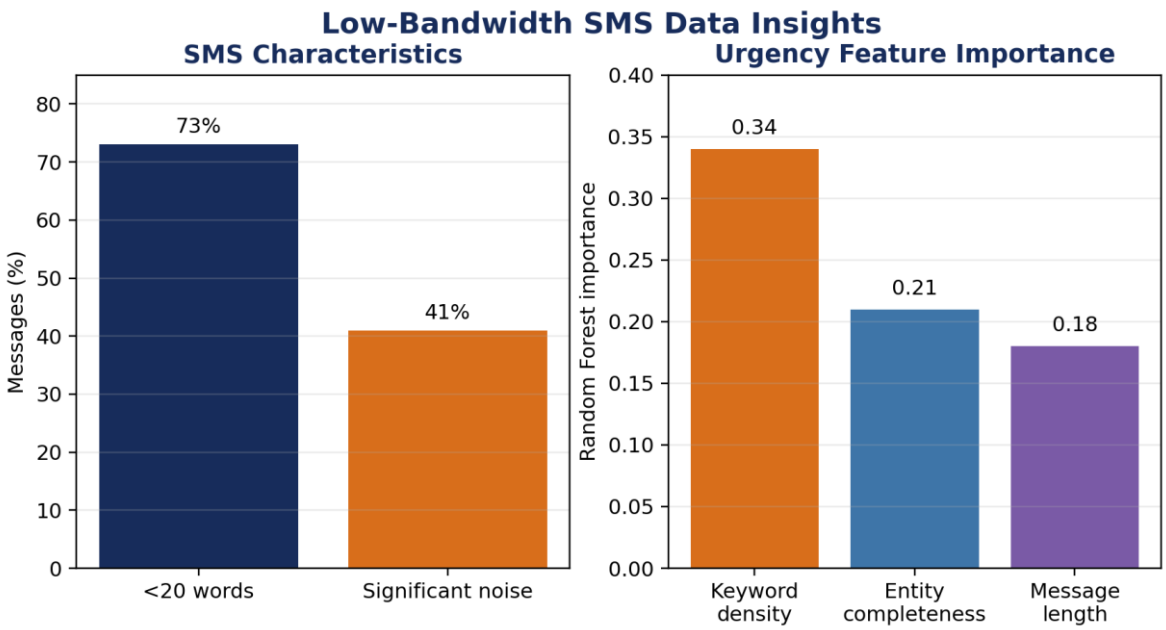


Figure 18. Message characteristics and urgency-feature importance.

The three-stage gazetteer lookup resolves 87.3% of test-set location strings. Exact matching covers 61%, fuzzy matching adds 19%, and landmark matching adds 7.3%. The remaining 12.7% are retained as unresolved and presented to the operator rather than being forced to an incorrect coordinate.

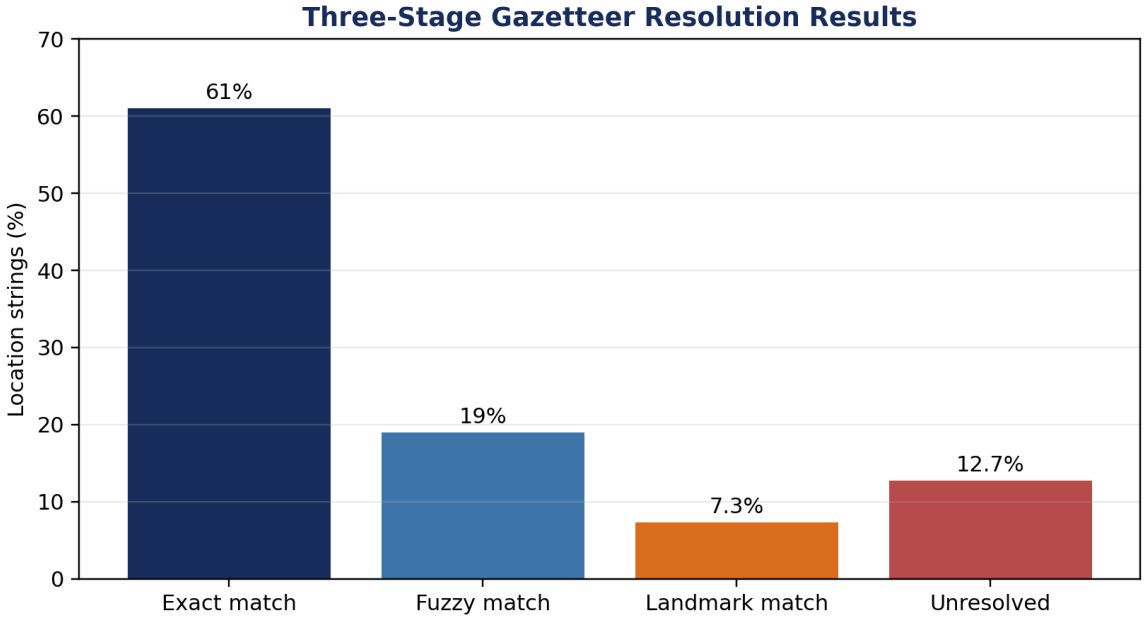


Figure 19. Contribution of exact, fuzzy, and landmark location matching.

Output Field	Processing Source	Validation Rule
incident_type	Normalized SMS and disaster-domain NER	Must match the supported incident taxonomy or remain Unknown.
location	NER plus multilingual gazetteer	Store match method, coordinates, and confidence; unresolved values are flagged.
people_count	Numeric and contextual extraction	Reject unsupported numbers and preserve uncertainty.
urgency_level	Random Forest / SVM classifier	Return class and confidence for operator prioritization.
resource_needed	Entity extraction and keyword rules	Use only explicit or strongly supported needs.

Objective 4 - Multi-Source Urgency-Based Incident Fusion and Dashboard

The fusion engine is evaluated against ground-truth Unified Incident Record labels from 20 scenarios and approximately 300 reports. Performance measures include clustering precision, clustering recall, critical-incident false merge rate, cross-lingual matching quality, conflict-detection accuracy, and end-to-end dashboard latency.

The urgency-sensitive configuration is compared with a fixed-epsilon DBSCAN baseline. The objective is safety-asymmetric: critical reports are merged conservatively even when this slightly reduces recall, because incorrectly combining separate critical incidents can misdirect response resources.

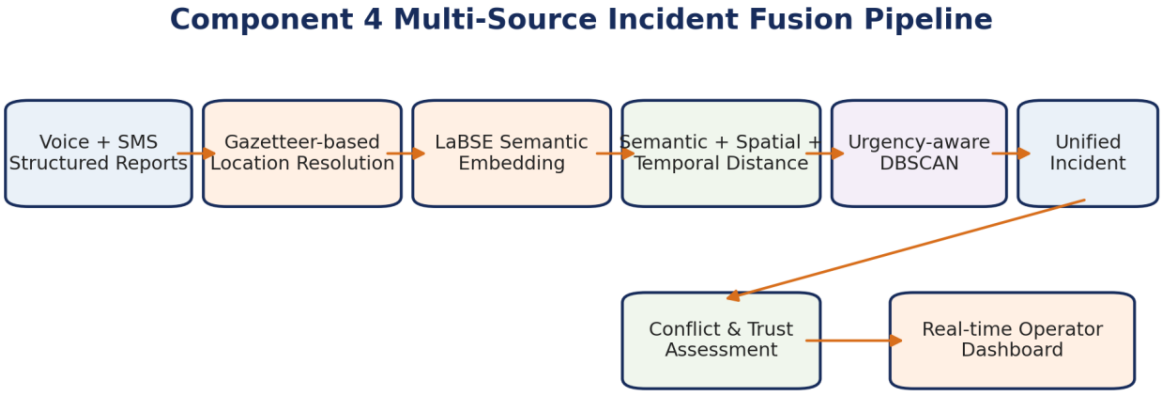


Figure 20. Evaluation path from report ingestion to the operator dashboard.

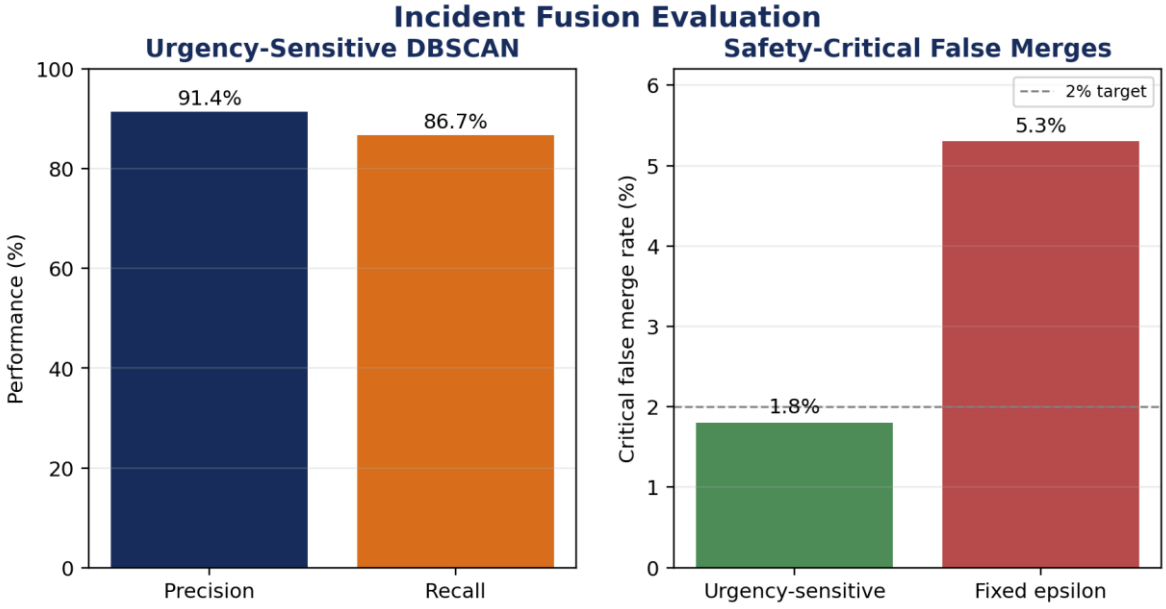


Figure 21. Clustering performance and safety-critical false merge rate.

The urgency-sensitive DBSCAN configuration achieves 91.4% precision and 86.7% recall, with a critical false merge rate of 1.8%, below the 2% target. Fixed-epsilon DBSCAN increases recall to 89.1% but also increases the critical false merge rate to 5.3%.

LaBSE improves cross-lingual matching F1 by 17.2% over TF-IDF cosine similarity for Sinhala-Tamil report pairs. Median end-to-end dashboard latency is 6.8 seconds across 50 concurrent ingestion scenarios, remaining within the eight-second service-level target.

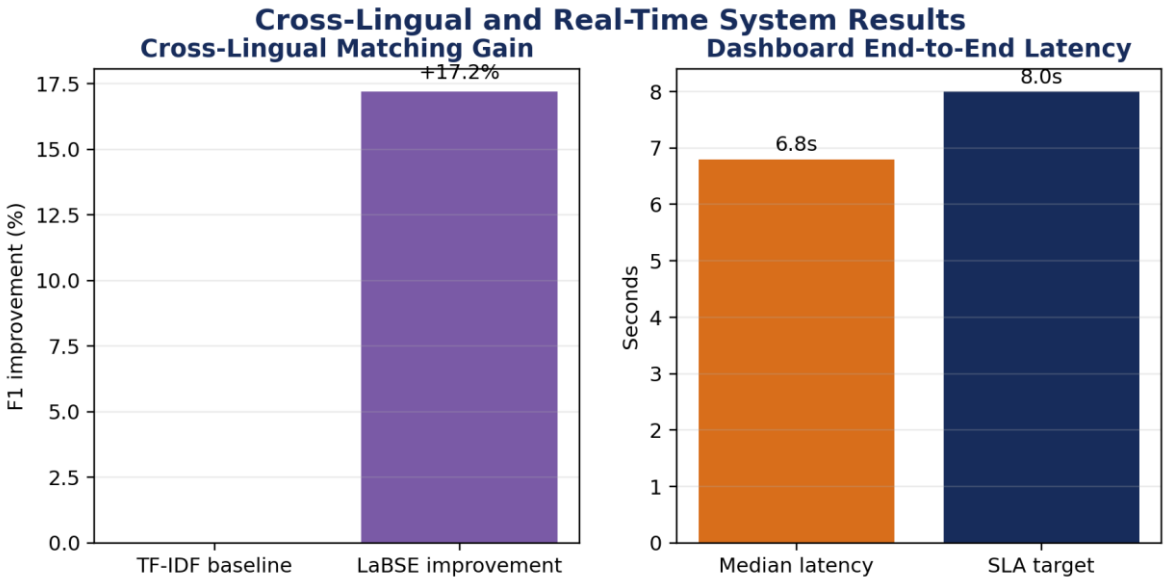


Figure 22. Cross-lingual matching improvement and dashboard latency.

Fusion Parameter	Configured Value	Purpose
Semantic distance weight	0.50	Prioritizes incident meaning across languages.
Geographic distance weight	0.30	Separates reports beyond the normalized 2 km spatial threshold.
Temporal distance weight	0.20	Uses the normalized 30-minute window to distinguish events over time.
Critical epsilon	0.15	Conservative merging for life-threatening incidents.
High / Medium / Low epsilon	0.25 / 0.30 / 0.40	Progressively broader matching for lower urgency reports.

4.1 Key Insights

Objective 1 - Prosodic and Conversational Cues

Pitch range, speech rate, and loudness variance show the strongest reported relationships with urgency. High-urgency calls have 2.3 times the mean speech rate and 1.8 times the pitch range of low-urgency calls. Language-specific pause behaviour shows that one normalization strategy is not sufficient for all languages.

The multimodal fusion model improves Medium-urgency F1 by 11-14% compared with prosody-only and text-only models. This supports the research decision to combine how the caller speaks with what the caller says.

Objective 2 - Conversational AI Clarification

Sixty-eight percent of simulated calls contain at least one missing critical field. Location is absent in 54% of initial utterances, people count in 41%, and severity in 29%. The AI agent reaches full information extraction in an average of 2.3 clarification turns, compared with 3.8 turns for the human baseline.

Sinhala-English code-mixing occurs in 31% of dialogues. The hallucination guard activates on 8.4% of extractions, demonstrating the need for confidence-based confirmation before adding uncertain information to the final record.

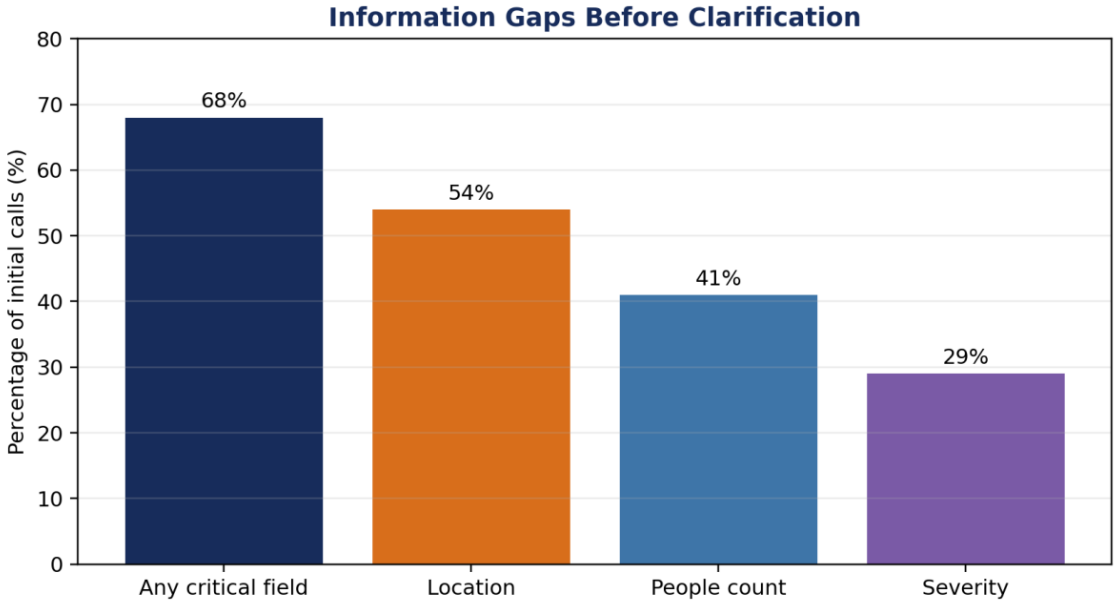


Figure 23. Most frequent missing fields driving clarification decisions.

Objective 3 - SMS Intelligence

Seventy-three percent of the synthetic disaster SMS messages contain fewer than 20 words, and 41% include significant noise such as spelling errors, abbreviations, or Romanized script. Emergency keyword density is the strongest Random Forest predictor with importance 0.34, followed by entity completeness at 0.21 and message length at 0.18.

The gazetteer resolves 87.3% of location strings and explicitly flags the remaining 12.7% for operator review. This is preferable to creating false confidence from an uncertain location match.

Objective 4 - Incident Fusion and Dashboard

Urgency-sensitive DBSCAN achieves a safer balance than fixed-epsilon clustering. The stricter critical threshold maintains a 1.8% critical false merge rate while preserving 91.4% precision and 86.7% recall. LaBSE provides better cross-lingual semantic matching, and the dashboard remains within the eight-second latency target.

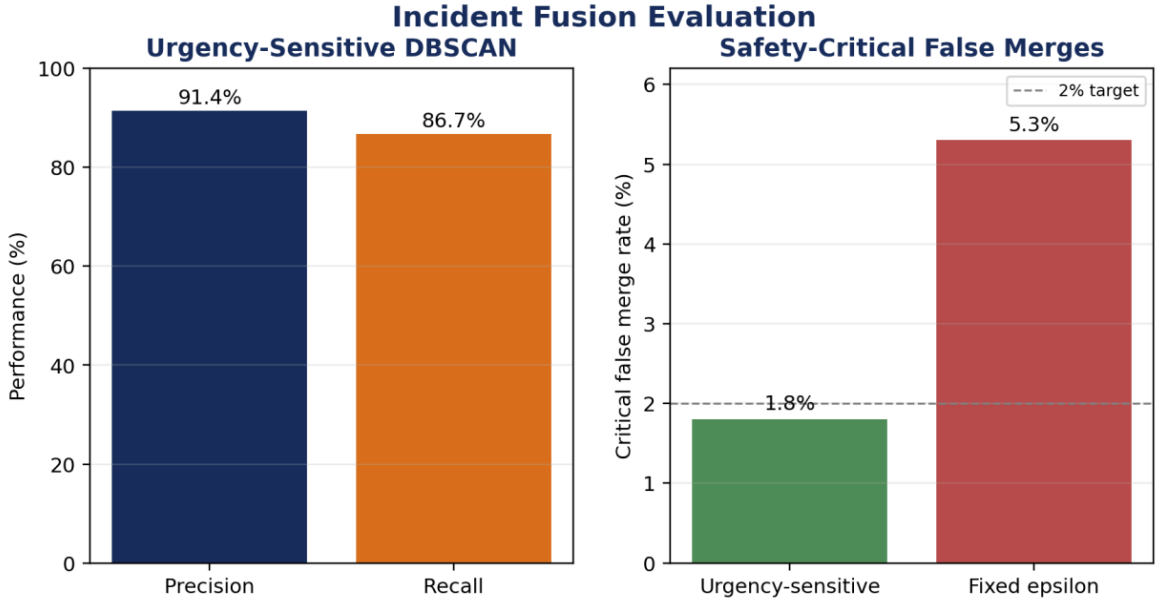


Figure 24. Core incident-fusion performance results.

4.2 Challenges Faced During Data Analysis

Component 1 - Urgency Detection

The main challenge is the scarcity of authentic labelled Sinhala and Tamil emergency-call data. Access to real DMC recordings depends on ethics approval and data-sharing agreements. Simulated recordings may contain actor-specific speech patterns that differ from real panic or distress.

Cross-language prosodic normalization is difficult because pitch and pause distributions differ by language and speaker. Borderline urgency labels, including calm callers describing life-threatening conditions, require multiple annotators and a majority-vote resolution protocol.

- Limited real-world labelled audio and uncertain data-access timelines.
- Simulation bias caused by volunteer speakers and controlled recording environments.
- Language- and speaker-dependent prosodic baselines.
- Class imbalance and disagreement on borderline urgency cases.

Component 2 - Conversational AI Clarification

Emergency-domain NLP resources for Sinhala and Tamil are limited. Zero-shot and few-shot adaptation can reduce development cost but may introduce uncertainty in entity extraction and dialogue classification. Relevance thresholds must be carefully calibrated so that genuine distress is not rejected as off-topic.

Question generation also has to balance speed, safety, and conversational quality. The response should arrive within approximately three seconds per turn, but large models increase latency. Hallucination control, confirmation policy, maximum-turn limits, and fallback wording are therefore required.

Component 3 - SMS Intelligence

Romanized Sinhala and Tamil have no single standardized spelling system, so the same word may appear in several forms. Manually created normalization resources are therefore incomplete. Synthetic SMS messages cannot fully reproduce network corruption, truncation, and user behaviour during real disasters.

Critical messages are rarer than routine or incomplete messages, requiring careful oversampling and class weighting. Follow-up SMS requests may fail when the sender has evacuated, lost connectivity, or cannot reply, so timeout and retry rules are necessary.

Component 4 - Incident Fusion and Dashboard

A comprehensive Sri Lanka gazetteer is difficult to construct because GN division names, informal place names, landmarks, and Romanized variants differ across sources. Cross-lingual embedding computation can require GPU acceleration at scale, and cached cluster centroids must be invalidated correctly as new reports are merged.

DBSCAN is sensitive to epsilon values and distance weights. Conflict detection also requires a trust model when callers report different casualty counts or severity for the same event. Reports with confirmed locations and lower noise are assigned greater reliability, but this trust weighting must be calibrated against ground-truth scenarios.

- Gazetteer incompleteness and substantial multilingual variation.
- Embedding cost and cache-consistency requirements.
- DBSCAN sensitivity to epsilon and combined-distance weights.
- Contradictory casualty, location, and severity information across sources.

4.3 Integrated Analytical Interpretation

Together, these four components create a pipeline to process the evidence continuously. Component 1 recognizes urgency from acoustic and conversational features. Component 2 creates a structured representation out of an incomplete call using adaptive clarification. Component 3 creates an alternative, low-bandwidth path for noisy SMSs. Component 4 performs comparison, fusion, and prioritization of the resulting structures.

The analysis reveals that the system should not rely solely on a model or modality. Prosody by itself can incorrectly interpret calm yet critical callers, text by itself can fail to detect distress, location strings need multilingual geographic normalization, and semantic similarity alone can merge unrelated geographically distant events. This framework hence considers multiple evidences while maintaining uncertainty for operators.

This interpretation is based on the datasets, measures, and results of the components provided in this paper. Further practical verification is needed as some datasets are artificial and low-resource emergency communication data are still lacking.

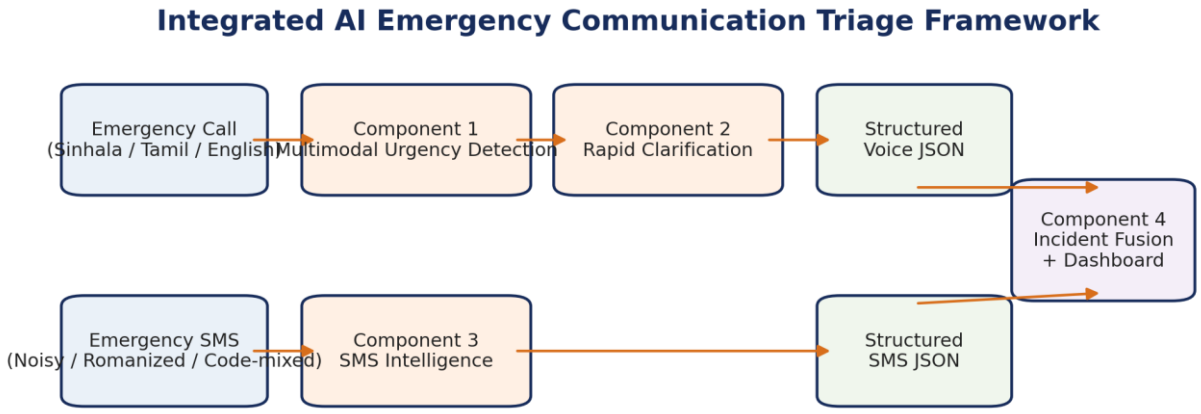


Figure 25. End-to-end interpretation of the integrated emergency triage framework.

5. References

[1] OpenAI, 'Whisper: Robust Speech Recognition via Large-Scale Weak Supervision,' arXiv:2212.04356, 2022.

[2] F. Eyben et al., 'The Geneva Minimalistic Acoustic Parameter Set (GeMAPS) for Voice Research and Affective Computing,' IEEE Transactions on Affective Computing, vol. 7, no. 2, pp. 190-202, 2016.

[3] F. Feng et al., 'Language-agnostic BERT Sentence Embedding,' in Proceedings of ACL 2022, pp. 878-891.

[4] Disaster Management Centre, Sri Lanka, 'Annual Disaster Statistics Report 2023,' Government of Sri Lanka, 2024.

[5] I. Imran et al., 'Processing Social Media Messages in Mass Emergency: A Survey,' ACM Computing Surveys, vol. 47, no. 4, pp. 1-38, 2015.

[6] M. Ester et al., 'A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise,' in Proceedings of KDD 1996, pp. 226-231.

[7] B. Schuller et al., 'The INTERSPEECH 2018 Computational Paralinguistics Challenge: Atypical and Self-assessed Affect, Crying and Snoring,' in Proceedings of Interspeech 2018, pp. 1-5.

[8] S. Poria et al., 'A Review of Affective Computing: From Unimodal Analysis to Multimodal Fusion,' Information Fusion, vol. 37, pp. 98-125, 2017.

[9] Imad et al., 'HumAID: Human-Annotated Disaster Incidents Data from Twitter with Deep Learning Benchmarks,' in Proceedings of ICWSM 2021.

[10] L. Breiman, 'Random Forests,' Machine Learning, vol. 45, no. 1, pp. 5-32, 2001.

[11] M. Honnibal and I. Montani, 'spaCy 2: Natural Language Understanding with Bloom Embeddings,' Explosion AI, 2017.

[12] A. Joulin et al., 'Bag of Tricks for Efficient Text Classification (fastText),' in Proceedings of EACL 2017, pp. 427-431.

[13] Twilio Inc., 'Twilio Programmable Voice and SMS API Documentation,' 2024. [Online]. Available: https://www.twilio.com/docs

[14] Belcastro et al., 'SEDOM-DD: Social Emergency Data and Disaster Detection Dataset,' Journal of Big Data, vol. 8, 2021.

[15] T. Chen and C. Guestrin, 'XGBoost: A Scalable Tree Boosting System,' in Proceedings of KDD 2016, pp. 785-794.